

Mathis BENCHIKH

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PROFILE: Graduate engineering student pursuing the Arts et Métiers–Cranfield double degree and an MSc in Aerospace Vehicle Design, seeking a **4–6-month final-year internship starting in May 2027**. I aim to contribute to industrial aerospace R&D projects while deepening my expertise in design, numerical simulation and guidance and control.

EDUCATION

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| Cranfield University , Cranfield, United Kingdom | Sep 2026 – Sep 2027 |
| MSc Aerospace Vehicle Design — Double degree with Arts et Métiers (in progress) | |
| <ul style="list-style-type: none">- Coursework: aircraft systems, aerodynamics, stability and control, composite structures, certification and safety.- Project: hydrogen tank design for the Cobalt Blue 3 project, sponsored by Safran | |
| Arts et Métiers , Paris / Châlons-en-Champagne, France | Sep 2024 – Jun 2027 |
| Generalist Engineering Degree — Mechanical and Industrial Engineering (in progress) | |
| <ul style="list-style-type: none">- Ranking: National top 9% (110 / 1,200 students); average grade: 15.8/20.- Coursework: fluid and structural mechanics, thermodynamics, electrical engineering and numerical simulation.- Student association: Head of Theory and Integration — AMSPACE (Oct 2025 – Feb 2026). | |
| Scientific Preparatory Classes, Lycée Fauriel , Saint-Étienne, France | Sep 2022 – Jun 2024 |
| <ul style="list-style-type: none">- MPSI–MP*: Mathematics, Physics and Engineering Sciences — Top 8 of 47 in MP*.- Intensive preparation for competitive entrance examinations to France’s leading engineering schools. | |

PROFESSIONAL EXPERIENCE

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| CNRS@CREATE , Singapore | Jun 2026 – Aug 2026 |
| Research Internship — Trajectory Optimisation for Autonomous Drones | |
| Co-author: contributed to writing a research paper submitted to the Journal of Control, Automation and Electrical Systems, currently under review. | |
| <ul style="list-style-type: none">- Co-developed a variational planner (Euler–Lagrange) and a collision-avoidance correction algorithm with the PIMM laboratory in a high-level international research environment.- Integrated the perception–guidance–control pipeline in Python: closed-loop trajectory tracking at approximately 20 Hz, validated on an experimental rover platform.- Team validation: 40/40 goals reached without collisions across two repeated scenarios; simulation–experiment comparison and analysis of tracking accuracy and safety margins. | |
| Emitech Group — LEFAE , Saint-Chamond, France | Jun 2025 – Jul 2025 |
| Shop-floor Internship — Aerospace Testing and Qualification | |
| <ul style="list-style-type: none">- Participated in qualification tests for Airbus A350F cargo-door seals: specimen preparation, stiffness testing and leak testing under pressure.- Instrumentation and data acquisition: thermocouples, pressure and displacement sensors; analysis of force–displacement curves.- Participated in four lithium-ion battery thermal-runaway propagation tests; prepared and monitored tests in accordance with safety and traceability protocols. | |
| ACADOMIA , Paris / Châlons-en-Champagne, France | Jan 2025 – Jun 2026 |
| Private Tutor — Mathematics, Physics and Chemistry | |
| <ul style="list-style-type: none">- Provided tailored support to 10 secondary-school students: adapted explanations to individual needs and monitored progress.- Developed students’ scientific reasoning, study methods and independent learning skills. | |

SKILLS AND INTERESTS

Technical skills: CAD: CATIA V5, Fusion 360, 3DEXPERIENCE | Simulation: STAR-CCM+, Abaqus | Programming: Python, MATLAB/Simulink.

Personal strengths: Scientific rigour, technical writing, communication and teamwork.

Languages: French (native), English (C1), Spanish (B1), Tamazight.

Interests: Hiking, travelling, football (R2 level), kickboxing, amateur esports.